

[Hanzhi Zhang]

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RESEARCH INTERESTS

Efficient AI, Hardware-Aware ML, Accelerated LLM Inference, HPC

EDUCATION

University of North Texas (UNT) Aug. 2023 - present
PhD Candidate, Computer Science
Advisors: Dr. Yunhe Feng (primary), Dr. Kewei Sha (co-advisor)

University of Birmingham (UoB) Sept. 2021 - Dec. 2022
MS, Data Science with **Distinction**

Xiamen University (XMU) Sept. 2017 - Aug. 2021
BS, Computer Science and Technology (**Honours**)

RESEARCH EXPERIENCE

Responsible AI Lab Denton, TX
Graduate Research Assistant, Responsible AI Aug. 2023 - present

- ◇ Research efficient LLM inference via **hardware-aware optimization of transformer attention**, designing **mixed-precision (FP16/INT8) mechanisms with sparsity and quantization**, implemented and evaluated on large-scale GPU clusters including **Polaris and TACC HPC systems**.

Argonne Training Program on Extreme-Scale Computing (ATPESC) Jun. 2024

- ◇ Selected as a participant in ATPESC 2024, an intensive program on extreme-scale computing covering GPU computing, parallel programming models, and performance optimization on leadership-class supercomputers.

OPPO, Inc. Shenzhen, China
Storage Backend Intern, File System Aug. 2020 - Apr. 2021

- ◇ Contributed to backend development of the **CubeFS distributed file system**, working on storage components including metadata management and storage optimization for large-scale cloud infrastructure.

SELECTED PROJECTS

Hardware-Aware Quantization (ongoing)

- ◇ Developing hardware-aware quantization for LLM inference by estimating weight importance and scheduling mixed-precision computation without retraining.
- ◇ Investigating sensitivity-aware quantization strategies that balance model accuracy and hardware efficiency for large transformer models.

TileMix: Tile-Centric Mixed Precision Attention for LLM Inference

- ◇ Proposed a **tile-level mixed-precision attention framework** enabling selective FP16/INT8 computation within FlashAttention-style kernels without retraining.
- ◇ Introduced a **compact precision policy** and scalable grouping mechanism supporting long-context inference with grouped-query attention and INT8 KV cache.
- ◇ Designed kernel-friendly precision scheduling to maintain compatibility with efficient attention implementations.

DAM: Dynamic Attention Mask for Long-Context LLM Inference

- ◇ Proposed a **dynamic sparse attention framework** assigning adaptive masks per attention map to preserve heterogeneous patterns across heads and layers.
- ◇ Fine-tuning-free approach that generalizes to varying input lengths and reduces attention computation while maintaining performance comparable to full attention.
- ◇ Enables efficient inference by dynamically identifying and preserving important attention regions.

PUBLICATIONS

- ◇ **Hanzhi Zhang**, Qiao Zhang, Qinglei Cao, Heng Fan, Yan Huang, Kewei Sha, Yunhe Feng. TileMix: Tile-Centric Mixed-Precision Attention for LLM Inference Acceleration. *ICML under review*, 2026.
- ◇ **Hanzhi Zhang**, Heng Fan, Kewei Sha, Yan Huang, Yunhe Feng. DAM: Dynamic Attention Mask for Long-Context Large Language Model Inference Acceleration. *ACL Findings*, 2025.
- ◇ **Hanzhi Zhang**, Kewei Sha, Heng Fan, Song Fu, and Yunhe Feng. D2SO: Detecting Distant and Small Objects for Vision-based Vehicle Autonomous Systems. *IEEE MOST*, 2025.
- ◇ **Hanzhi Zhang**, Sumera Anjum, Heng Fan, Weijian Zheng, Yan Huang, Yunhe Feng. Poly-FEVER: A Multilingual Fact Verification Benchmark for Hallucination Detection in Large Language Models. *ICML under review, 2026, arXiv preprint*
- ◇ Bing Fan, Minghao Li, **Hanzhi Zhang**, Shaohua Dong, Naga Prudhvi Mareedu, Weishi Shi, Yunhe Feng, Yan Huang, Heng Fan. Towards Visual Query Segmentation in the Wild. *ECCV under review, 2026*
- ◇ Mingchen Li, **Hanzhi Zhang**, Heng Fan, Junhua Ding, Yunhe Feng. Harmful Factuality Hallucination: LLMs Correcting What They Shouldn't. *EACL Findings*, 2026
- ◇ Sumera Anjum, **Hanzhi Zhang**, Wenjun Zhou, Eun Jin Paek, Xiaopeng Zhao, Yunhe Feng. HALO: Hallucination Analysis and Learning Optimization to Empower LLMs with Retrieval-Augmented Context for Guided Clinical Decision Making. *IEEE/ACM CHASE*, 2025.
- ◇ Tong Xie, **Hanzhi Zhang**, Shaozhou Wang, Yuwei Wan, Imran Razzak, Chunyu Kit, Wenjie Zhang, Bram Hoex. ByteScience: Bridging Unstructured Scientific Literature and Structured Data with Auto Fine-tuned Large Language Model in Token Granularity. *IEEE ICDMW*, 2024.

TALKS, AWARDS & HONORS

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| ◇ Poster Presentation, CSE Open House, University of North Texas | 2023–2026 |
| ◇ Invited Talk: Responsible AI, AIxHEART | Oct 2025 |
| ◇ Poster Presentation, University Research Day, UNT | 2024, 2025 |
| ◇ IEEE MOST Student Travel Grant Award | Apr 2025 |
| ◇ First Place Award, UNT Engineering Showcase Poster Presentation | Mar 2024 |
| ◇ Research Presentation, Responsible AI Lab AI Seminar | 2024–2026 |

OPEN SOURCE CONTRIBUTIONS

- ◇ **[DAM]**: Dynamic sparse attention masking for long-context LLM inference.
- ◇ **[Poly-FEVER]**: Multilingual fact verification dataset for hallucination detection in large language models.
- ◇ **[CubeFS]**: Distributed file system.